AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XI

ANNUAL EXAMINATION (THEORY) 2023

Computer Science Paper II

Time: 1 hours 30 minutes Marks: 35

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INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's Signature**

RUBRIC

2. There are EIGHT questions. Answer ALL questions. Question 7 and 8 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

   Read each question carefully.
   Use a black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid, or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a scientific calculator if you wish.

Q.1. (Total 6 Marks)

All devices that contain embedded, specialised computers are called computing devices.

Identify any TWELVE computing devices that are used in daily life.

Q.2. (Total 3 Marks)

i. Describe the process of writing data on an optical disk. (2 Marks)

ii. Write ONE advantage of optical storage over magnetic storage. (1 Mark)

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Q.3. (Total 3 Marks)

Write any THREE differences between Complex Instruction Set Computer (CISC) and Reduced Instruction Set Computer (RISC) architectures.

Q.4. (Total 3 Marks)

USB port is a plug-and-play port.

i. Describe the meaning of plug-and-play. (1 Mark)

ii. Describe the response of a computer when it detects a device connected via a USB port. (1 Mark)

iii. List the names of any TWO devices, other than USB flash drive, that can be connected with a computer via a USB port. (1 Mark)

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Q.5. (Total 3 Marks)

Differentiate between infra-red and bluetooth based on the given characteristics.

Characteristics	Infra-red	Bluetooth
Bandwidth		
Range		
Number of Devices		

Q.6. (Total 3 Marks)

Differentiate between primary key and candidate key.

Primary Key	Candidate Key

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Q.7. (Total 7 Marks)

EITHER

a.

i. Define the term ‘communication device’. (1 Mark)

ii. Describe the following communication devices: (6 Marks)

I. Hub
II. Switch
III. Gateway

OR

b.

i. Define global positioning system (GPS). (1 Mark)

ii. Describe the components of GPS. (4 Marks)

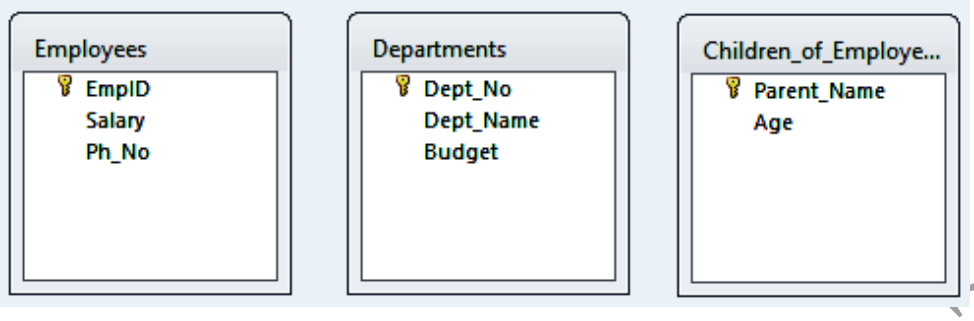
ii. Mention any TWO applications of GPS. (2 Marks)

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Q.8. (Total 7 Marks)

EITHER

a. Consider the given image showing three tables in an MS Access database of a company.



A number of employees work in each department of the company. Each department is managed by an employee. Each child of an employee is identified by the employee's name. Furthermore, the company has the policy that only one parent can work for the company.

Draw an entity-relationship diagram (ERD) for this database. (7 Marks)

Space for ERD

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OR

Consider the given table named as Professors.

Prof_ID	First_Name	Last_Name	Age	Experience_in_Years	Specialisation
P101	Dilawar	Khan	47	22	Accounting
P102	Sana	Ahmed	52	30	Marketing
P103	Faisal	Arbab	45	23	Finance

b. Write MS Access SQL queries to

i. create this table with appropriate data types and with the following field lengths. Set **Prof_ID** as primary key. (3 Marks)

Field Name	Field Length
Prof_ID	5
First_Name	20
Last_Name	15
Specialisation	25

ii. insert last record shown in the table. (1 Mark)

iii. remove **Age** field from the table. (1 Mark)

iv. add **Gender** field in the table having field length of 6 characters. (1 Mark)

v. change the field length of **Specialisation** so that it should accept up to 30 characters. (1 Mark)

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